How Rich Are Mixture Attitudes?

Theory and Experimental Evidence

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Oxford

October 9th, 2026

Mixtures in economics

Consumption
bundles of goods

Risk
money lotteries

Uncertainty
acts over states

Time
consumption streams

Social
allocations

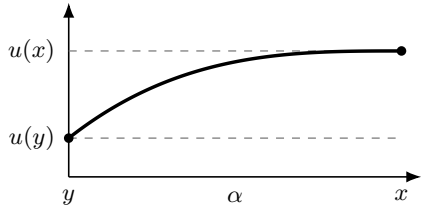
Strategy
mixed strategies

Mixtures in economics

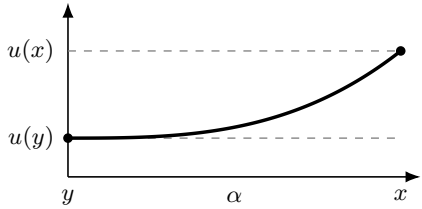
Consumption bundles of goods	**Risk** money lotteries	**Uncertainty** acts over states
Time consumption streams	**Social** allocations	**Strategy** mixed strategies

Most economic environments combine objects through a *mixture operation*

Mixture operation describes preferences

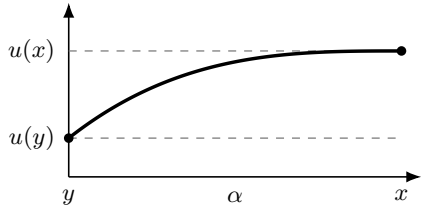


Mixture loving

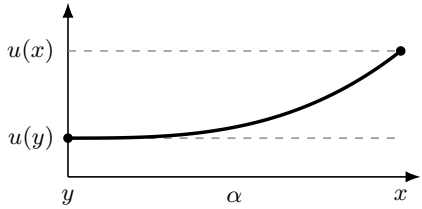


Mixture averse

Mixture operation describes preferences



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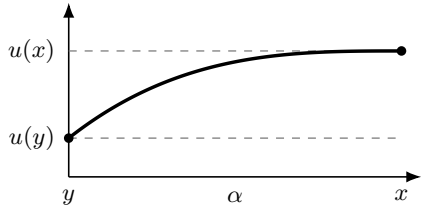
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Most theories assume one-sided attitudes

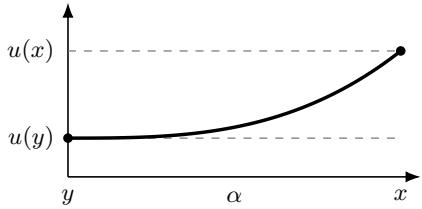
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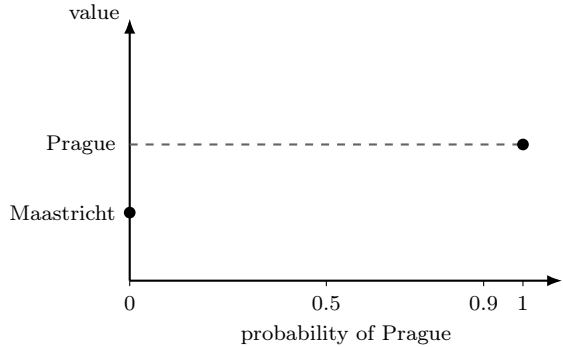
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This paper: what about preferences that are neither?

Our interest: richer mixture attitudes

Example: surprise trips

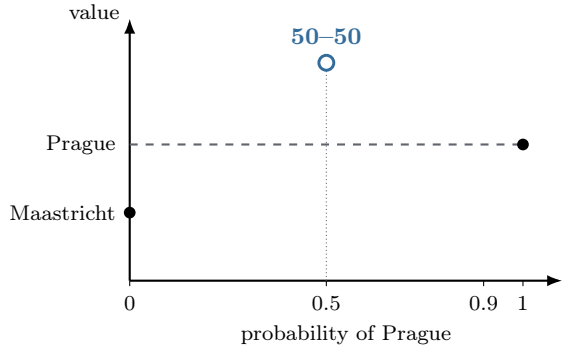
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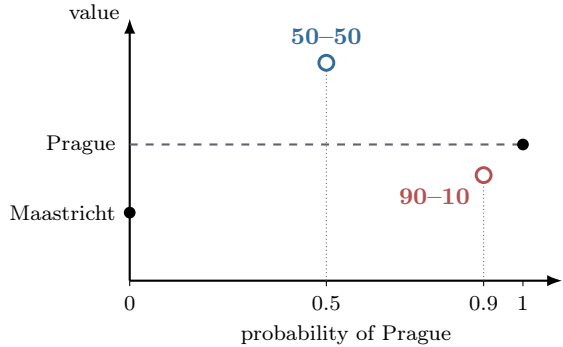
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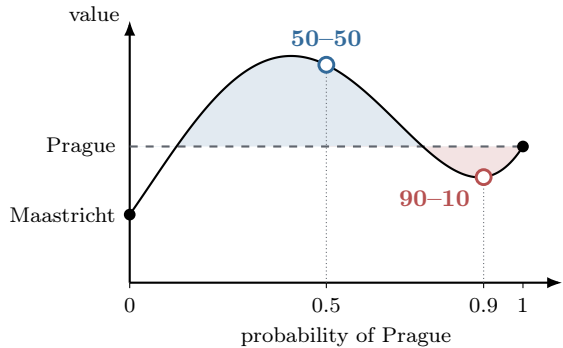
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Like to randomize if sufficiently random, else prefer certainty

Why richer mixture attitudes?

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Theory: No conceptual/organizing framework for mixed attitudes

We build a hierarchy to study and classify mixed attitudes, revealing new insights

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Theory: No conceptual/organizing framework for mixed attitudes

We build a hierarchy to study and classify mixed attitudes, revealing new insights

Empirics: No evidence or tests of relevance/prevalence of mixed attitudes

Show mixed attitudes are common and missed in a previously well-studied setting

Theory

Framework

- **Objects:** compact convex $\mathcal{X} \subseteq [0,1]^d$ containing the diagonal $(t, \ldots, t)$
- **Mixing:** $x \alpha y = \alpha x + (1-\alpha) y$

Rich environment: lotteries, Anscombe–Aumann acts, consumption bundles, social allocations, etc.

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- **Preferences** $\succsim$: complete, transitive, mixture continuous, strictly monotone
  - Irrational, incomplete, or stochastic choice can imply strange mixture attitudes.
  - Focus: intentional, rational choice.

A hierarchy of mixture attitudes: MMA

Definition (Mixed mixture attitudes, MMA)

For some $x^+ \succsim y^+$, $x^- \precsim y^-$ and $\alpha, \beta \in (0,1)$,

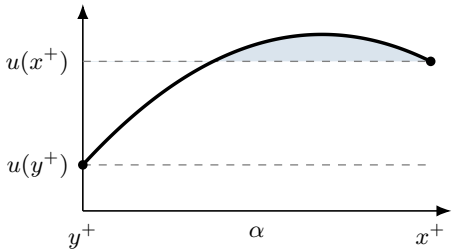
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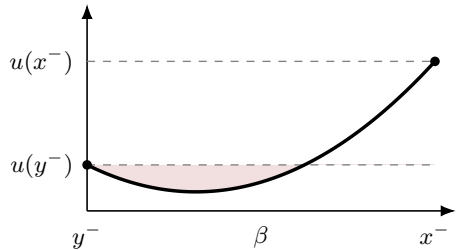
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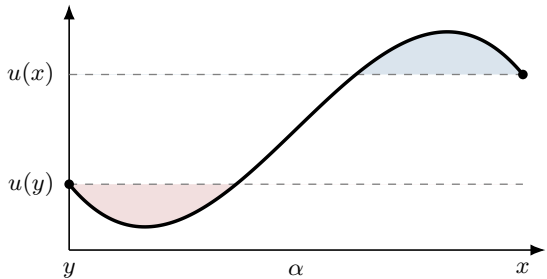
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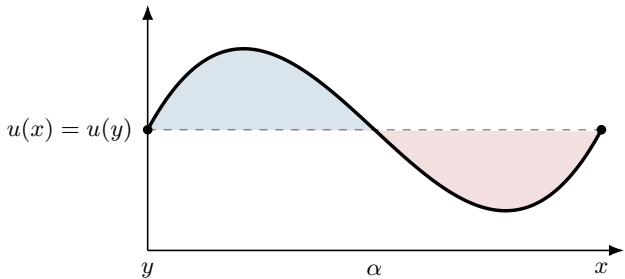
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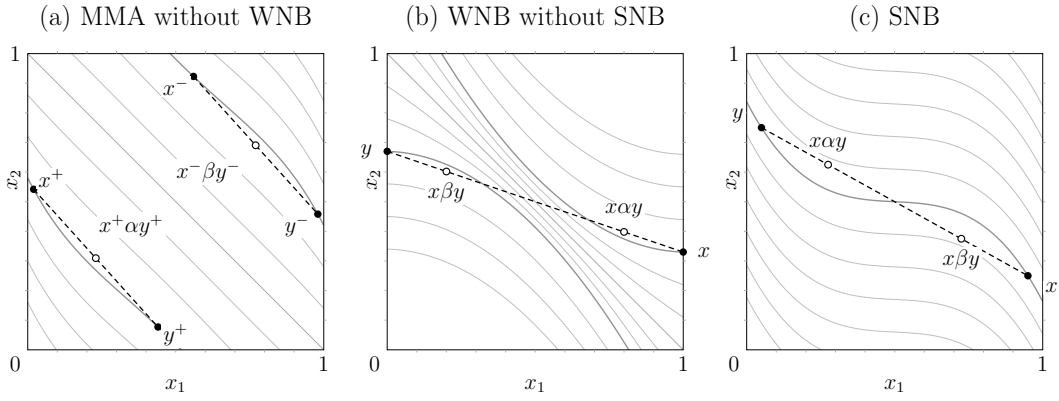
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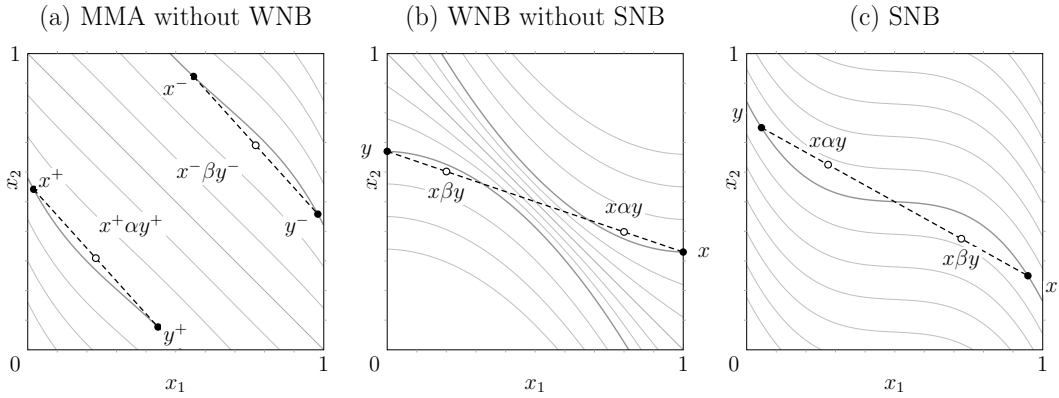


Three example indifference maps



MMA $\supsetneq$ WNB $\supsetneq$ SNB.

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Natural question: is SNB knife-edge/pathological?

SNB is generic

Calibrated utility: $U_{\succsim}(x) = t$ if $x \sim (t, \ldots, t)$
Distance between preferences: $d(\succsim, \succsim') = \|U_{\succsim} - U_{\succsim'}\|_\infty$

Theorem (Genericity)

SNB preferences are open and dense in MMA preferences.

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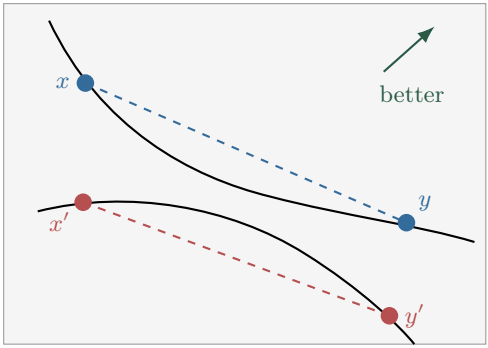
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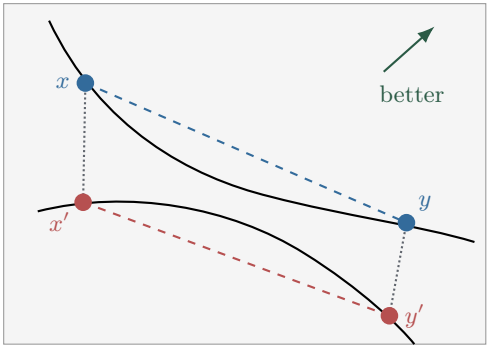
The result is robust: it also holds in the $C^1, \ldots, C^\infty$ topologies.

Why SNB is generic: proof sketch



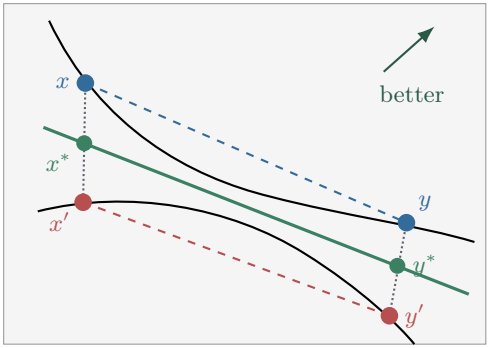
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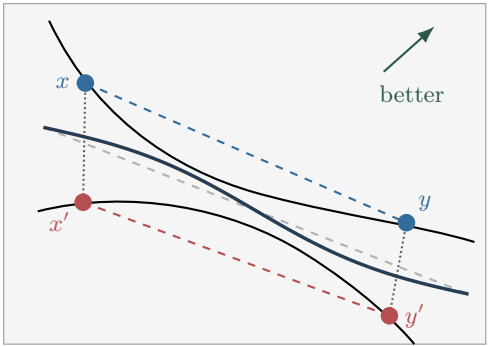
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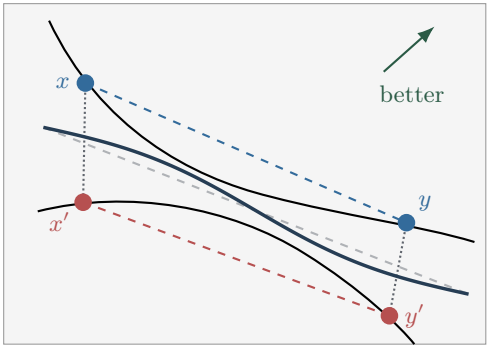
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Exact neutrality is the knife edge. But what if a model builds it in?

Classifying models

Uncertainty	MMA, not WNB	WNB, not SNB	SNB
α-maxmin EU			
Choquet EU			
Invariant biseparable			

These models build in mixture neutrality in a structured way.

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▸ *Precise statement*

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Risk	MMA, not WNB	WNB, not SNB	SNB
Quadratic utility	✓	–	–
Mixture symmetry	✓	✓	–
RDEU, $n \geq 4$	–	–	✓
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- Masatlioglu and Raymond (2016): Quadratic utility $\cap$ RDEU$\neq \emptyset$
- We show: only if mixture attitudes are globally one-sided

Literature audit

Existing evidence lacking - design methodology cannot identify mixed attitudes

Domain	*N*	**Weak Loving**	**Weak Averse**	**Strict Loving**	**Strict Averse**	**MMA**	**WNB**	**SNB**
Randomization and lotteries	52	**46** (88%)	**46** (88%)	**20** (38%)	**9** (17%)	**6** (12%)	**3** (6%)	**3** (6%)
Consumption budgets	19	**12** (63%)	**12** (63%)	**9** (47%)	**2** (11%)	**2** (11%)	**2** (11%)	**2** (11%)
Time preferences	83	**72** (87%)	**72** (87%)	**8** (10%)	**3** (4%)	**3** (4%)	0	0
Portfolio and asset choice	66	**58** (88%)	**58** (88%)	**1** (2%)	0	0	0	0
Charitable giving	100	**88** (88%)	**88** (88%)	**7** (7%)	**2** (2%)	**2** (2%)	**2** (2%)	**2** (2%)
Effort allocation	55	**27** (49%)	**27** (49%)	**1** (2%)	**1** (2%)	**1** (2%)	0	0
Social and interpersonal allocation	169	**154** (91%)	**154** (91%)	**25** (15%)	**9** (5%)	**5** (3%)	**1** (1%)	**1** (1%)

This table documents whether, in principle, papers can identify various attitudes

Why does mixed mixture attitude arise?

- Theory is explicity *general* and *descriptive* - *why* may be domain dependent
- Experiment focuses on a specific environment to discuss particular channels

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- Theory is explicity *general* and *descriptive* - *why* may be domain dependent
- Experiment focuses on a specific environment to discuss particular channels
- General takeaway: rich mixture attitudes arise out of rich tradeoffs/considerations

Experiment

The experiment: what we want to know

How to identify, experimentally, mixed mixture attitudes?

Are mixed mixture attitudes, and SNB in particular, common?

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How to identify, experimentally, mixed mixture attitudes?

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Environment: randomization

A sure prize vs. a lottery over prizes

Familiar environment - extensively studied

Set-up: eliciting rankings

- 83 participants
- **Real**(istic) Prizes: four-day trips to 10 European destinations
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Intentional choice/evaluation of a familiar high-stakes object

Choices: sure trips vs. surprise lotteries

After ranking the trips, $x_1 \succ x_2 \succ \cdots \succ x_{10}$, face binary choices, including:

- x_1 vs. $p\,x_1 + (1-p)\,y$ and y vs. $p\,x_1 + (1-p)\,y$
- $p \in \{0.1,\, 0.5,\, 0.9\}$
- $y \in \{x_2,\, x_3,\, x_5,\, x_{10}\}$
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Each mixture is observed at three weights, against both endpoints.

Testing SNB

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Definition (SNB*)

For some $x \gtrsim y$ and $\alpha < \beta$ in $(0,1)$,

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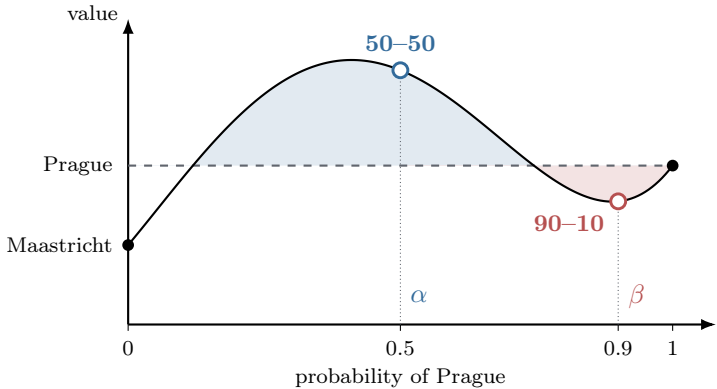
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Proposition

Under the basic axioms, SNB $\iff$ *SNB**.

Graphical illustration



$$\alpha < \beta \;\Rightarrow\; \text{SNB}^* \iff \text{SNB}$$

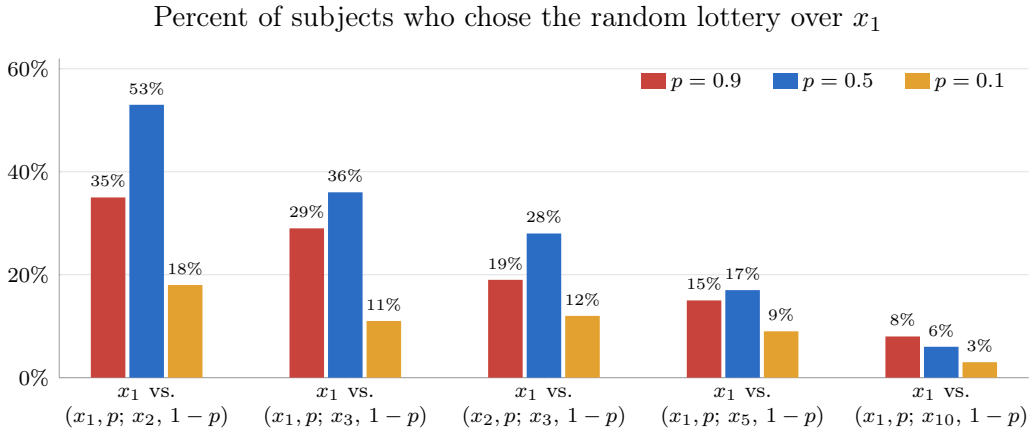
Result: SNB is common

- **41%** of participants (34 of 83) display SNB at least once

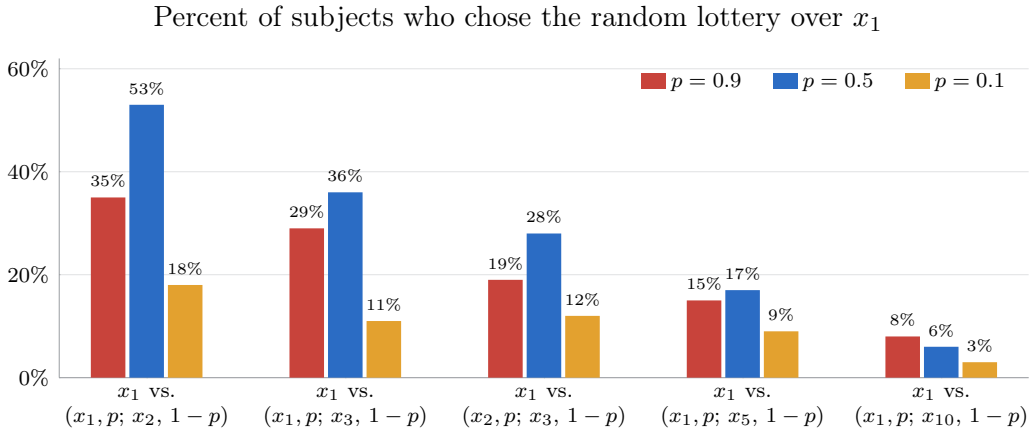
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- Mixed attitudes almost always take the form of SNB

Choosing the lottery over x_1



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Note: read one-dimensionally, 50–50 $\succ$ Prague violates stochastic dominance.

▸ A mistake?

What drives SNB? Rich/Multiplicity of Motives

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Experiment

Suggestive evidence from elicited motives for choice and various other choices

Conclusion

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2. **Experimentally**, common but understudied

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- Contrast with complex low-stakes choices $\Rightarrow$ global mixture preferences

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Consequences for understanding optimization, complexity and aggregation.

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1. **Theoretically**, rich taxonomy, not knife-edge
2. **Experimentally**, common but understudied

Mixture attitudes capture the discrepancy between preference and mixture operation.

- Richness of attitude captures deliberate choice + richness in environment
- Contrast with complex low-stakes choices $\Rightarrow$ global mixture preferences

Consequences for understanding optimization, complexity and aggregation.

Many other domains/models of study left

Appendix

When neutrality is built in

Definition (Neutral benchmark)

$n \in \mathcal{X}$ is a neutral benchmark if, for all $x, y \in \mathcal{X}$ and $\gamma \in (0,1)$,

$$x \succsim y \iff x\gamma n \succsim y\gamma n.$$

Proposition

If $\succsim$ *has a neutral benchmark* $n \in \mathrm{ri}(\mathcal{X})$*, then*

$$\succsim \textit{ displays MMA} \iff \succsim \textit{ displays SNB.}$$

Certainty independence makes every constant act a neutral benchmark:
for α-maxmin, Choquet and invariant biseparable preferences, MMA $\iff$ SNB.

Back

A mistake?

Read alone, 50–50 $\succ$ Prague violates first-order stochastic dominance in the participant's own ranking (Nielsen and Rehbeck, 2022). But these choices are:

Costly	$\approx$**10%** of a trip's value given up to keep the surprise
Systematic	**53%** choose the surprise at 50–50, but only **35%** at 90–10
Valued	**54%** of lottery builders pay $\geq$ £5 to delay the reveal
Specific to trips	**4%** choose the 50–50 lottery when prizes are money

Trips are bundles. All cost £420, so a cheaper city means a better hotel. Without attribute-wise dominance, monotonicity is silent. Money is the one-attribute limit.

Back

A choice from our experiment

Sure trip

your favorite destination

(say, Prague)

vs.

50–50 surprise
Prague or your 2nd favorite
(say, Maastricht)

53%
choose the surprise

90–10 surprise
90% Prague, 10% Maastricht

35%
choose the surprise

Many make both choices: 50–50 surprise $\succ$ Prague $\succ$ 90–10 surprise